

PROJNAME

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Change history

1.0.0	07/10/2026	First release
1.0.0	08/10/2026	Fix R1234 value due to issue xyz etc etc etc. Update C14 to 10 nF to reduce roll-off frequency.
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD

System overview

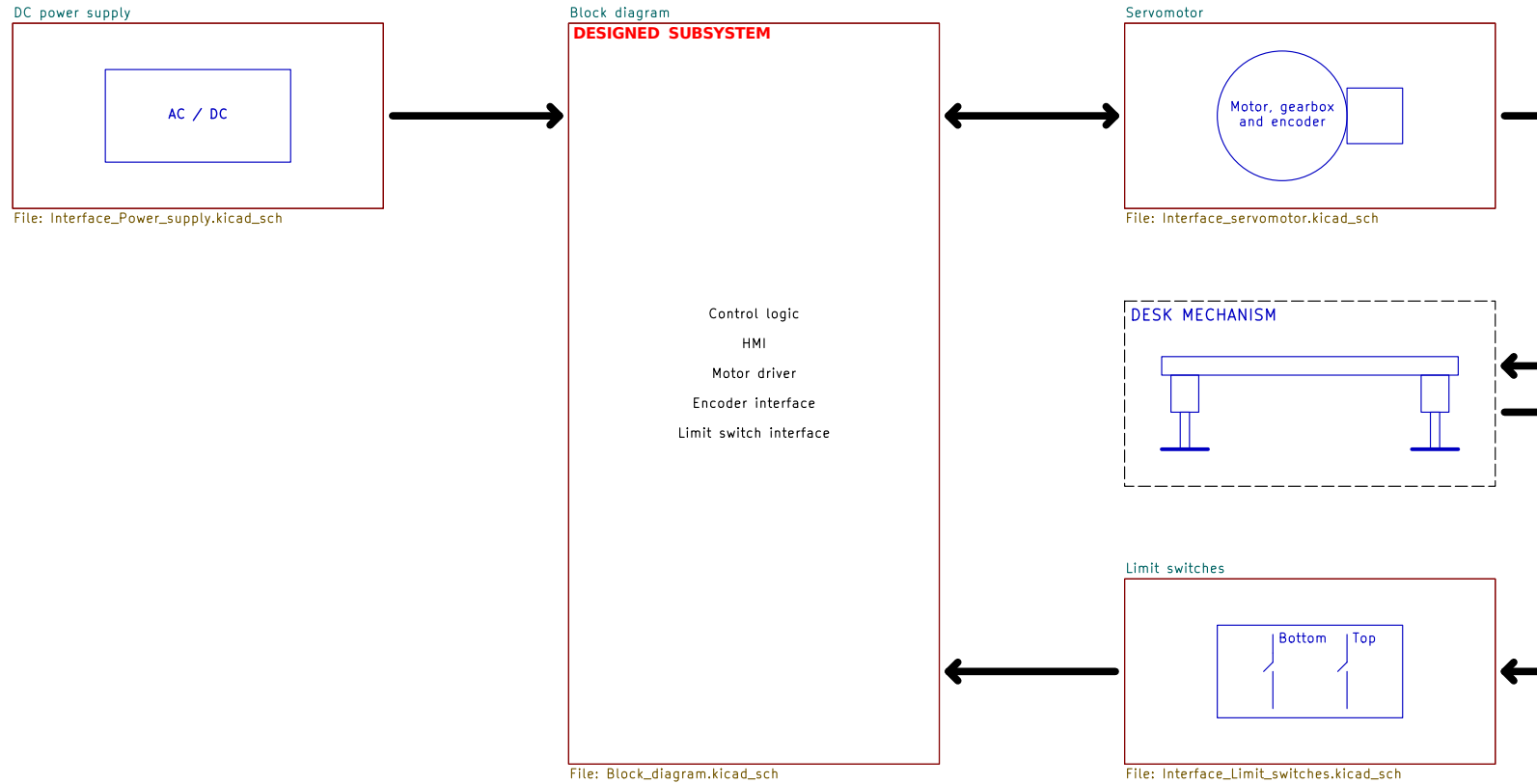


File: System_overview.kicad_sch



PROJNAME		Sheet: Root	
		File: ktest.kicad_sch	
Notes This is a comment 1 field - potentially not used This is a comment 2 field - potentially not used This is a comment 3 field - potentially not used This is a comment 4 field - potentially not used	Rev: 0.0.1	Drwn: FLR	
	Mat: Concept	Rev'd: FLR	
	Date: 2026-10-06	Rls'd: FLR	
	Dwg: NA	Size: A4	
		Generated with KiCad	Page: 1/11

System overview



This sheet shows the overall system and the interfaces to external components.
External components are shown for system context only and are not part of this schematic.
Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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PROJNAME

Notes

Sheet: System overview

File: System_overview.kicad_sch

Rev:

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Dwg: NA

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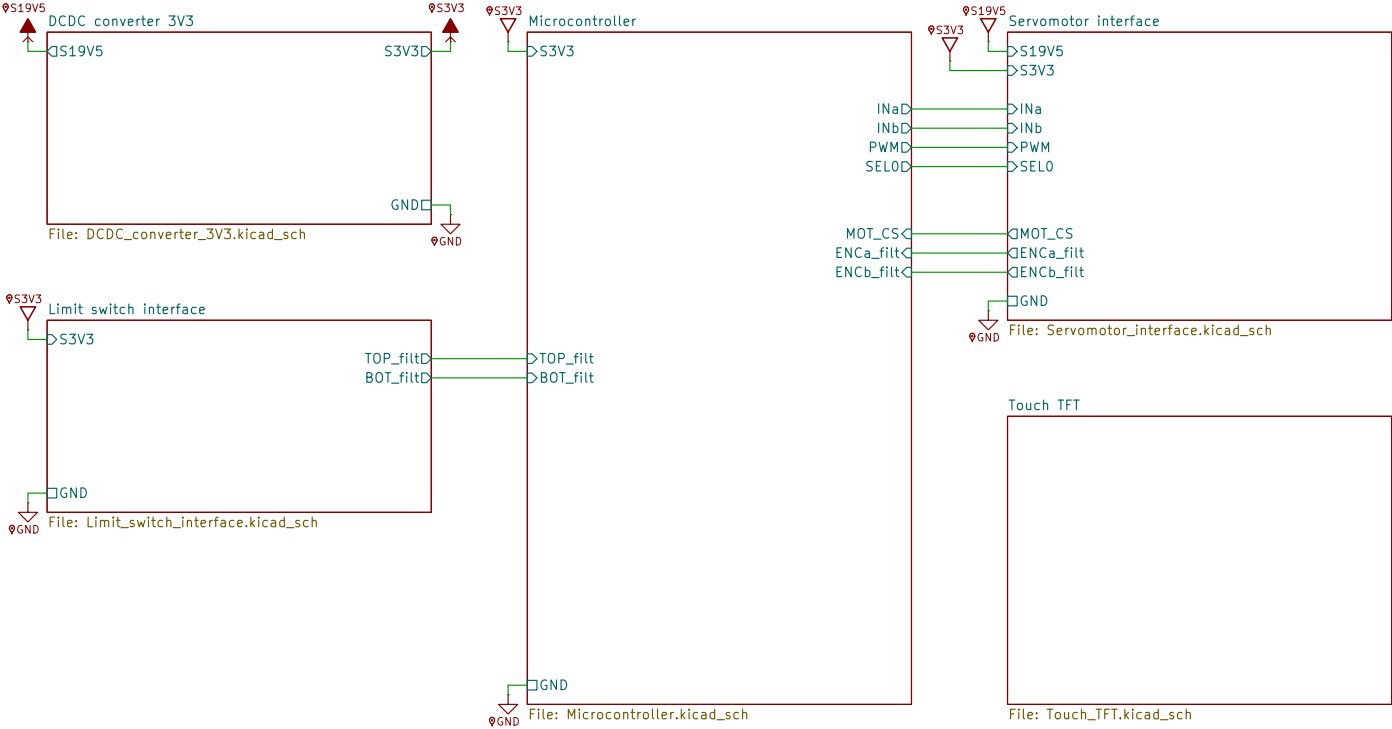
Rev'd: FLR

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Size: A4

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Block diagram



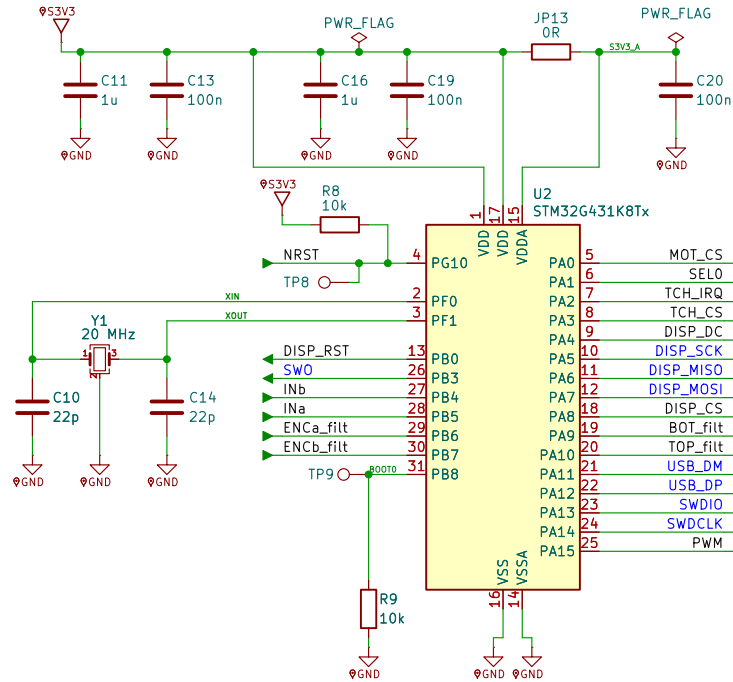
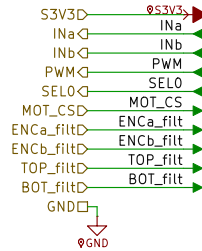
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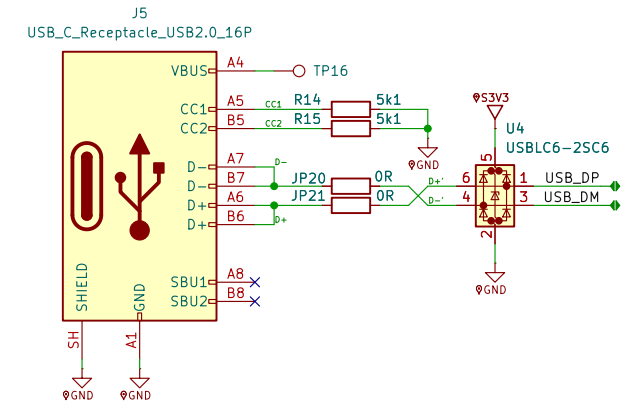
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Notes	Rev:		Drwn: FLR
	Mat: Concept		Rev'd: FLR
	Date:		RLs'd: FLR
	Dwg: NA		Size: A4
	Generated with KiCad		Page: 3/11

Microcontroller

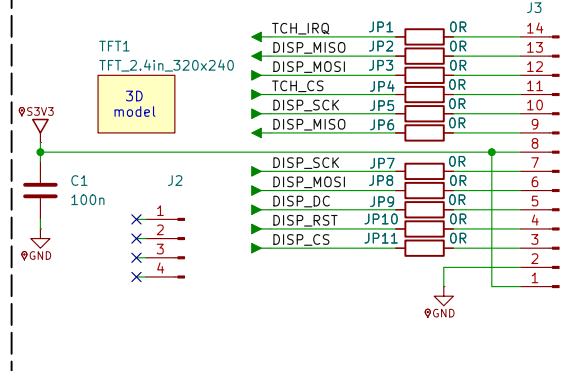
Block interface



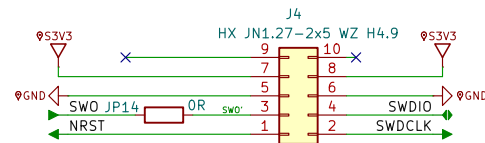
USB interface (serial comms port)



TFT display



SWD header



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Notes

Sheet: Microcontroller

File: Microcontroller.kicad_sch

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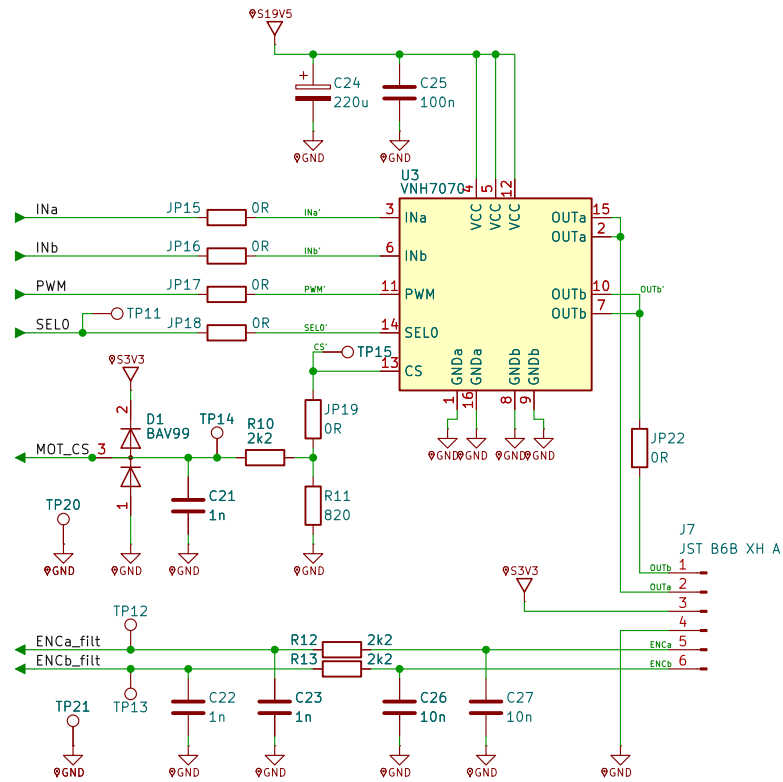
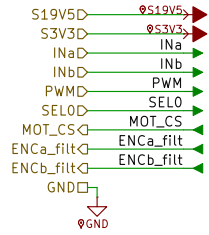
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Size: A4

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Servomotor interface

Block interface



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Notes

Sheet: Servomotor interface

File: Servomotor_interface.kicad_sch

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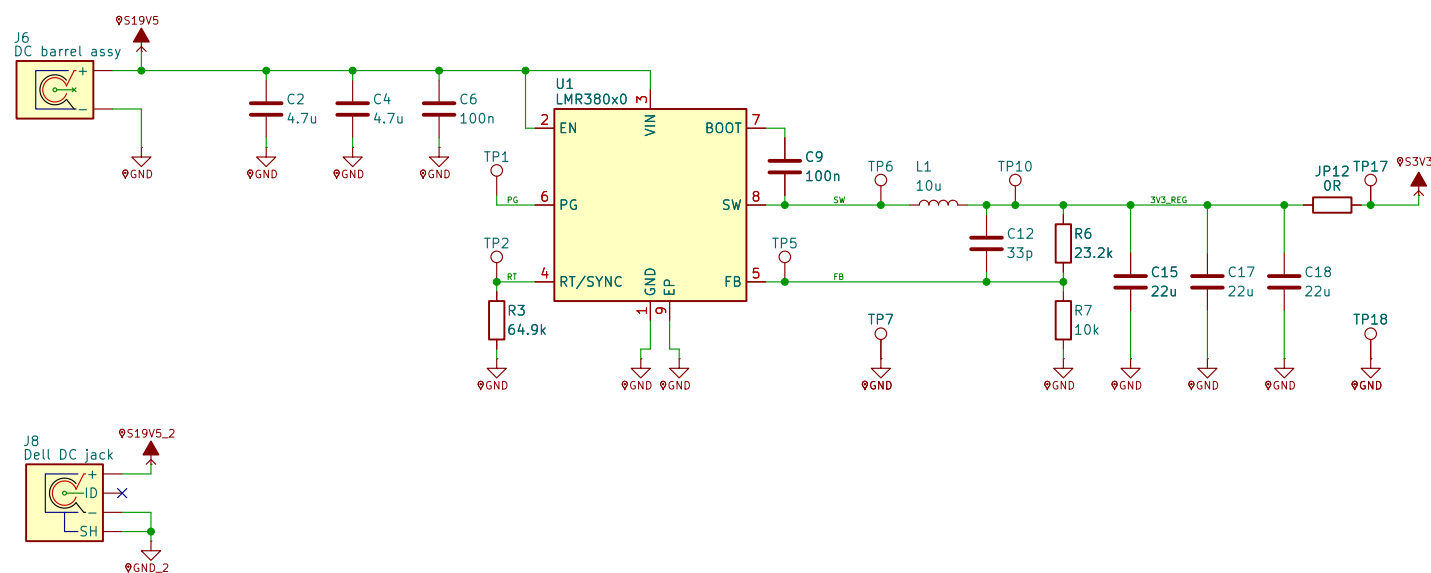
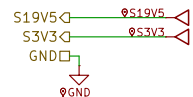
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DCDC converter 3V3

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Sheet: DCDC converter 3V3

File: DCDC converter 3V3.kicad sch

Rev:	
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Dwg: NA

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Rev'd: FLR

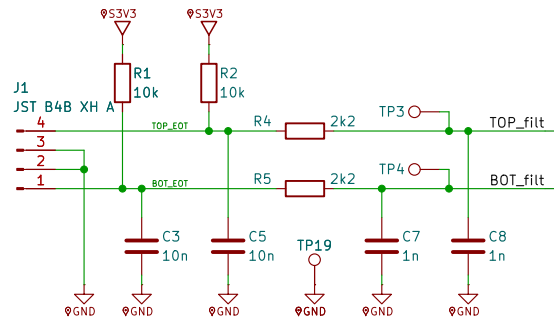
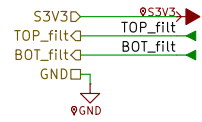
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Limit switch interface

Block interface



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Sheet: Limit switch interface	
File: Limit_switch_interface.kicad_sch	
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1	2	3	4	5	6
Block interface					Touch TFT
A					A
B					B
C					C
D					D
1	2	3	4	5	6



PROJNAME Notes	Sheet: Touch TFT File: Touch_TFT.kicad_sch	
	Rev:	Drwn: FLR
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DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

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Notes

Sheet: DC power supply	
File: Interface_Power_supply.kicad_sch	
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Servomotor

Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)

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Notes

Sheet: Servomotor

File: Interface_servomotor.kicad_sch

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Interface specification

JST B4B XH A / JST XHP-4 connector

- Pin Description
- 1 Bottom limit switch (Normally open, switch closes to GND)
 - 2 GND
 - 3 GND
 - 4 Top limit switch (Normally open, switch closes to GND)



PROJNAME	Sheet: Limit switches	
	File: Interface_Limit_switches.kicad_sch	
	Rev:	Drwn: FLR
	Mat: Concept	Rev'd: FLR
	Date:	RLs'd: FLR
Notes	Dwg: NA	Size: A4
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